Day 19 Assignment - 2

Description:

Learned different ways to handle asynchronous operations in Node.js using callbacks, Promises, and async/await.

This day focused on non-blocking file operations and modern async coding styles.

Challenge 7 - (Required Files - callback-demo.js, data.txt)

Used the fs module to write content into a file and read it back in the console.

This demonstrated basic file operations in Node.js.

Comand to run: ***node callback-demo.js***

JS callback-demo.js ✕ data.txt

JS callback-demo.js > ...

```javascript
const fs = require("fs");

fs.readFile("data.txt", "utf8", (err, data) => {
    if (err) {
        console.error("Error reading file:", err);
        return;
    }

    console.log("File Content:");
    console.log(data);

    setTimeout(() => {
        console.log("Read operation completed");
    }, 1000);
});
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS D:\Wipro\Assignments\Day 19\Day 19 Assignemnt-2> node callback-demo.js
File Content:
Node.js callback demo content
by Hemanth Saladi
Read operation completed
PS D:\Wipro\Assignments\Day 19\Day 19 Assignemnt-2>
```

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Challenge 8 - (Required Files – promise-demo.js, input.txt)

Copied file content from one file to another using Promise chaining with fs.promises.

This reduced callback nesting and improved readability.

Comand to run: ***node promise-demo.js***

promise-demo.js input.txt

promise-demo.js > ...

```javascript
const fs = require("fs").promises;

fs.readFile("input.txt", "utf8")
  .then((data) => {
    return fs.writeFile("output.txt", data);
  })
  .then(() => {
    console.log("File copied successfully!");
  })
  .catch((error) => {
    console.error("Error:", error);
  });
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS D:\Wipro\Assignments\Day 19\Day 19 Assignemnt-2> node .\promise-demo.js
File copied successfully!
PS D:\Wipro\Assignments\Day 19\Day 19 Assignemnt-2>
```

promise-demo.js input.txt

input.txt

```
This content will be copied using Promises.
Hello Wipro and Great Learning.
```

output.txt

output.txt

```
This content will be copied using Promises.
Hello Wipro and Great Learning.
```

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Challenge 9 – (Files – events-demo.js)

Performed the same file copy operation using async/await with try/catch for cleaner syntax.

This showed a modern and structured way of handling asynchronous tasks.

Comand to run: **node *async-demo.js***

JS async-demo.js ✕ | async-output.txt

JS async-demo.js > ...

```javascript
const fs = require("fs").promises;

async function copyFile() {
    try {
        const data = await fs.readFile("input.txt", "utf8");

        await new Promise((resolve) => setTimeout(resolve, 1000));

        await fs.writeFile("async-output.txt", data);

        console.log("File copied successfully using async/await!");
    } catch (error) {
        console.error("Error:", error);
    }
}

copyFile();
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS D:\Wipro\Assignments\Day 19\Day 19 Assignemnt-2> node async-demo.js
File copied successfully using async/await!
PS D:\Wipro\Assignments\Day 19\Day 19 Assignemnt-2>
```

JS async-demo.js | async-output.txt ✕

async-output.txt

```
This content will be copied using Promises.
Hello Wipro and Great Learning.
```

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